



```
$ learn-linux --by-doing
```

Linux Commands Interactive Guide

Clear explanations. Real examples. Safer troubleshooting.

CLICKABLE NAVIGATION

Expanded and redesigned from the supplied Linux command reference

Contents

Choose a topic. Every section contains copyable commands, a plain-language explanation, guidance on when to use it, and safety notes where the command changes the system.

01 Basic navigation	02 Files and directories
03 Searching	04 Permissions and ownership
05 Users and groups	06 Processes and jobs
07 System information	08 Disk and storage
09 Memory and CPU	10 Networking
11 Package management	12 Services and systemd
13 Archives and compression	14 Text processing
15 Redirection and pipes	16 Environment variables
17 SSH and remote copy	18 Log troubleshooting
19 Cron scheduling	20 DevOps troubleshooting
21 Useful one-liners	22 Interview essentials
23 Scenario playbook	

How to read a command

command	The program or shell built-in to run.
-options	Flags that change behavior. Short options often begin with one hyphen; long options usually begin with two.
arguments	The file, directory, user, host, pattern, or value the command acts on.
 > >>	Shell operators: pass output onward, overwrite a file, or append to a file.

A safe practice loop

1. Inspect first. 2. Run the least destructive command. 3. Read the output. 4. Change one thing. 5. Verify the result. Use a test directory or disposable virtual machine while learning.

1

Basic navigation

8 command examples in this section

pwd

Shows the current working directory.

```
$ pwd
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

ls

Lists files and directories.

```
$ ls
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

ls -l

Lists files with details like permissions, owner, size, and date.

```
$ ls -l
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

ls -la

Lists all files, including hidden files.

```
$ ls -la
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

cd

Changes directory.

```
$ cd /var/log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

cd ..

Moves one directory back.

```
$ cd ..
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

cd ~

Goes to the user's home directory.

```
$ cd ~
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

clear

Clears the terminal screen.

```
$ clear
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

2

Files and directories

17 command examples in this section

touch

Creates an empty file.

```
$ touch test.txt
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

mkdir

Creates a new directory.

```
$ mkdir devops
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

mkdir -p

Creates parent directories if they do not exist.

```
$ mkdir -p app/logs/backup
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-p: create parents or include process information

cp

Copies a file.

```
$ cp file1.txt file2.txt
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

cp -r

Copies a directory recursively.

```
$ cp -r app backup-app
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-r: recursive or reverse, depending on the command

mv

Renames a file.

```
$ mv old.txt new.txt
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

mv

Moves a file to another location.

```
$ mv app.log /tmp/
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

rm

Deletes a file.

```
$ rm test.txt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

rm -r

Deletes a directory recursively.

```
$ rm -r old-folder
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

OPTION CLUE

-r: recursive or reverse, depending on the command

rm -rf

Forcefully deletes a directory. Use carefully.

```
$ rm -rf old-folder
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

OPTION CLUE

-r: recursive or reverse, depending on the command

SAFETY

DANGER: recursive forced deletion is immediate. Print and verify the exact path first.

cat

Displays file content.

```
$ cat app.log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

less

Opens large files page by page.

```
$ less /var/log/syslog
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

head

Shows the first 10 lines of a file.

```
$ head app.log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

head -n

Shows the first 20 lines.

```
$ head -n 20 app.log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-n: numeric output or a requested count

tail

Shows the last 10 lines of a file.

```
$ tail app.log
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

tail -f

Shows live log updates.

```
$ tail -f app.log
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

tail -n

Shows the last 100 lines.

```
$ tail -n 100 app.log
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-n: numeric output or a requested count

3

Searching

10 command examples in this section

grep

Searches for text inside a file.

```
$ grep "error" app.log
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

grep -i

Searches text without case sensitivity.

```
$ grep -i "error" app.log
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE**-i**: ignore case or use an identity file, depending on the command**grep -r**

Searches recursively inside directories.

```
$ grep -r "database" /etc/
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE**-r**: recursive or reverse, depending on the command**grep -n**

Shows matching line numbers.

```
$ grep -n "failed" app.log
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE**-n**: numeric output or a requested count**find**

Finds files by name.

```
$ find / -name nginx.conf
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE**-n**: numeric output or a requested count**find**

Finds all files. .log

```
$ find /var/log -name "*.log"
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE**-n**: numeric output or a requested count**find -mtime**

Finds files modified in the last 1 day.

```
$ find /var/log -mtime -1
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

find -size

Finds files larger than 100 MB.

```
$ find / -type f -size +100M
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

locate

Quickly finds files by name.

```
$ locate nginx.conf
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

updatedb

Updates the locate command database.

```
$ sudo updatedb
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

4

Permissions and ownership

7 command examples in this section

chmod

Changes file permissions.

```
$ chmod 755 script.sh
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Caution: this changes system state. Confirm the path, device, or permissions before running it.

chmod +x

Gives execute permission to a script.

```
$ chmod +x deploy.sh
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

chmod 644

Gives the owner read/write access and everyone else read-only access.

```
$ chmod 644 file.txt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Caution: this changes system state. Confirm the path, device, or permissions before running it.

chown

Changes the owner of a file.

```
$ sudo chown rahul file.txt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

chown user:group

Changes both the owning user and group.

```
$ sudo chown rahul:devops file.txt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

chown -R

Changes ownership throughout a directory tree.

```
$ sudo chown -R rahul:devops /app
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

OPTION CLUE

-R: recursive

SAFETY

Caution: this changes system state. Confirm the path, device, or permissions before running it.

ls -l

Displays permission bits, owner, group, size, and modification time.

```
$ ls -l script.sh
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

5

Users and groups

9 command examples in this section

whoami

Shows the current logged-in user.

```
$ whoami
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

id

Shows user ID and group details.

```
$ id
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

useradd

Creates a new user.

```
$ sudo useradd devuser
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

useradd -m

Creates a user with home directory.

```
$ sudo useradd -m devuser
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

passwd

Sets or changes user password.

```
$ sudo passwd devuser
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

usermod -aG

Adds user to a group.

```
$ sudo usermod -aG sudo devuser
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

OPTION CLUE

-a: include all or preserve metadata, depending on the command

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

groupadd

Creates a new group.

```
$ sudo groupadd devops
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

su

Switches to another user.

```
$ su - devuser
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

sudo

Runs a command with root privileges.

```
$ sudo systemctl restart nginx
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

6

Processes and jobs

12 command examples in this section

ps

Shows current shell processes.

```
$ ps
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

ps aux

Shows all running processes.

```
$ ps aux
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

with grep ps

Finds a specific running process.

```
$ ps aux | grep nginx
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

top

Shows live CPU and memory usage.

```
$ top
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

htop

Interactive process viewer.

```
$ htop
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

kill

Kills a process using PID.

```
$ kill 1234
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

kill -9

Forcefully kills a process.

```
$ kill -9 1234
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Caution: SIGKILL prevents cleanup. Try a normal kill first and use -9 only when necessary.

pkill

Kills a process by name.

```
$ pkill nginx
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

jobs

Shows background jobs.

```
$ jobs
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

bg

Runs a stopped job in background.

```
$ bg
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

fg

Brings a background job to foreground.

```
$ fg
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

nohup

Runs a process even after logout.

```
$ nohup python3 app.py &
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

7

System information

7 command examples in this section

uname -a

Shows system and kernel information.

```
$ uname -a
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-a: include all or preserve metadata, depending on the command

hostname

Shows system hostname.

```
$ hostname
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

uptime

Shows how long the system has been running.

```
$ uptime
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

date

Shows current date and time.

```
$ date
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

cal

Shows calendar.

```
$ cal
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

who

Shows logged-in users.

```
$ who
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

w

Shows logged-in users and their activity.

```
$ w
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

8

Disk and storage

8 command examples in this section

df -h

Shows free and used space for mounted filesystems in readable units.

```
$ df -h
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

OPTION CLUE

-h: human-readable units

df -h /

Shows capacity for the filesystem that contains the root directory.

```
$ df -h /
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

OPTION CLUE

-h: human-readable units

du -sh

Calculates one human-readable total for a directory.

```
$ du -sh /var/log
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

du --max-depth

Shows the size of each immediate child directory.

```
$ du -h --max-depth=1 /var
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

OPTION CLUE

-h: human-readable units

lsblk

Shows disks and partitions.

```
$ lsblk
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

mount

Mounts a disk or partition.

```
$ sudo mount /dev/sdb1
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Caution: this changes system state. Confirm the path, device, or permissions before running it.

umount

Unmounts a mounted filesystem.

```
$ sudo umount /mnt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

SAFETY

Caution: this changes system state. Confirm the path, device, or permissions before running it.

fdisk -l

Lists disk partitions.

```
$ sudo fdisk -l
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

OPTION CLUE

-l: long/detailed listing

SAFETY

Caution: this changes system state. Confirm the path, device, or permissions before running it.

9

Memory and CPU

6 command examples in this section

free -h

Shows total, used, available, and cached memory in readable units.

```
$ free -h
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

OPTION CLUE

-h: human-readable units

vmstat

Prints CPU, memory, process, and I/O statistics every two seconds.

```
$ vmstat 2
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

lscpu

Displays CPU architecture, cores, threads, sockets, and virtualization details.

```
$ lscpu
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

top

Opens a live, sortable view of processes and system load.

```
$ top
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

ps by memory

Lists the processes using the most memory.

```
$ ps aux --sort=-%mem | head
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

ps by CPU

Lists the processes using the most CPU.

```
$ ps aux --sort=-%cpu | head
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

10

Networking

17 command examples in this section

ip addr

Shows IP addresses.

```
$ ip addr
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ip a

Short form of . ip addr

```
$ ip a
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ip route

Shows routing table.

```
$ ip route
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ping

Tests network connectivity.

```
$ ping google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ping -c

Sends only 4 ping packets.

```
$ ping -c 4 google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

curl

Sends HTTP request.

```
$ curl https://example.com
```

WHEN TO USE

Use to test a URL, inspect an HTTP response, or retrieve content without a browser.

curl -I

Shows only HTTP headers.

```
$ curl -I https://example.com
```

WHEN TO USE

Use to test a URL, inspect an HTTP response, or retrieve content without a browser.

curl API test

Requests an application health endpoint and prints the response body.

```
$ curl http://localhost:8080/health
```

WHEN TO USE

Use to test a URL, inspect an HTTP response, or retrieve content without a browser.

wget

Downloads a remote file to the current directory.

```
$ wget https://example.com/file.zip
```

WHEN TO USE

Use to test a URL, inspect an HTTP response, or retrieve content without a browser.

netstat

Lists listening TCP/UDP sockets and, when permitted, their processes.

```
$ netstat -tulnp
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ss

Uses the modern socket tool to list listening ports and processes.

```
$ ss -tulnp
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ss with grep

Filters socket output to investigate port 80.

```
$ ss -tulnp | grep :80
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

telnet

Attempts an interactive TCP connection to a host and port.

```
$ telnet 10.0.0.5 3306
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

nc

Probes a TCP port without sending application data.

```
$ nc -zv 10.0.0.5 3306
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

traceroute

Shows the network hops toward a destination.

```
$ traceroute google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

dig

Queries DNS and displays detailed answer, authority, and timing data.

```
$ dig google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

nslookup

Performs a quick DNS name or address lookup.

```
$ nslookup google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

11

Package management

10 command examples in this section

apt update

Updates package index.

```
$ sudo apt update
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

apt install

Installs a package.

```
$ sudo apt install nginx
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

apt remove

Removes a package.

```
$ sudo apt remove nginx
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

apt upgrade

Upgrades installed packages.

```
$ sudo apt upgrade
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

yum install

Installs a package.

```
$ sudo yum install nginx
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

yum remove

Removes a package.

```
$ sudo yum remove nginx
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

yum update

Updates packages.

```
$ sudo yum update
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

dnf install

Installs a package.

```
$ sudo dnf install nginx
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

dnf update

Updates packages.

```
$ sudo dnf update
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

dnf remove

Removes a package.

```
$ sudo dnf remove nginx
```

WHEN TO USE

Use the package manager that matches your Linux distribution.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

12

Services and systemd

10 command examples in this section

systemctl start

Starts a service now, without changing boot behavior.

```
$ sudo systemctl start nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

systemctl stop

Stops a running service now.

```
$ sudo systemctl stop nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

systemctl restart

Stops and starts a service so configuration changes take effect.

```
$ sudo systemctl restart nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

systemctl status

Shows current state, recent log lines, and the main process ID.

```
$ sudo systemctl status nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

systemctl enable

Configures a service to start automatically at boot.

```
$ sudo systemctl enable nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

systemctl disable

Removes automatic startup links; it does not stop the service now.

```
$ sudo systemctl disable nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

journalctl

Reads logs collected by systemd-journald.

```
$ journalctl
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

journalctl -u

Filters systemd logs to one service unit.

```
$ journalctl -u nginx
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

journalctl -u -f

Follows new log entries for one service in real time.

```
$ journalctl -u nginx -f
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

journalctl --since

Limits journal output to entries after a time expression.

```
$ journalctl --since today
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

13

Archives and compression

7 command examples in this section

tar -cvf

Creates a tar archive.

```
$ tar -cvf backup.tar /app
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

tar -xvf

Extracts a tar archive.

```
$ tar -xvf backup.tar
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

tar -czvf

Creates compressed tar archive.

```
$ tar -czvf backup.tar.gz /app
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

tar -xzvf

Extracts compressed tar archive.

```
$ tar -xzvf backup.tar.gz
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

zip

Compresses a file into zip format.

```
$ zip app.zip app.txt
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

zip -r

Compresses a directory.

```
$ zip -r app.zip /app
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

OPTION CLUE

-r: recursive or reverse, depending on the command

unzip

Extracts a zip file.

```
$ unzip app.zip
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

14

Text processing

11 command examples in this section

awk

Prints the first column from a file.

```
$ awk '{print $1}' access.log
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

awk

Prints first and third columns.

```
$ awk '{print $1, $3}' file.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

with awk df

Prints filesystem and usage percentage.

```
$ df -h | awk '{print $1, $5}'
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE

-h: human-readable units

sed

Replaces with in output. dev prod

```
$ sed 's/dev/prod/g' config.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

cut

Cuts and prints specific fields.

```
$ cut -d ':' -f1 /etc/passwd
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

sort

Sorts lines alphabetically.

```
$ sort names.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

sort -r

Sorts lines in reverse order.

```
$ sort -r names.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE

-r: recursive or reverse, depending on the command

uniq

Removes duplicate lines.

```
$ sort names.txt | uniq
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

uniq -c

Counts duplicate lines.

```
$ sort names.txt | uniq -c
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

WC

Counts lines, words, and characters.

```
$ wc file.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

wc -l

Counts only lines.

```
$ wc -l file.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE

-l: long/detailed listing

15

Redirection and pipes

3 command examples in this section

>

Writes output to a file and overwrites existing content.

```
$ echo "hello" > file.txt
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

2>

Redirects errors to a file.

```
$ ls /wrongpath 2>
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

&>

Redirects both output and error to a file.

```
$ command &> output.log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

16

Environment variables

5 command examples in this section

env

Shows all environment variables.

```
$ env
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

echo

Prints the value of a variable.

```
$ echo $PATH
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

export

Creates an environment variable.

```
$ export ENV=dev
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

export

Sets application-related environment variable.

```
$ export DB_HOST=localhost
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

unset

Removes an environment variable.

```
$ unset DB_HOST
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

17

SSH and remote copy

8 command examples in this section

ssh

Connects to a remote server.

```
$ ssh user@server-ip
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

ssh

Connects to server using username and IP.

```
$ ssh ubuntu@10.0.0.5
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

ssh -i

Connects using a private key.

```
$ ssh -i key.pem ubuntu@10.0.0.5
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

OPTION CLUE

-i: ignore case or use an identity file, depending on the command

scp

Copies file from local to remote server.

```
$ scp app.log ubuntu@10.0.0.5:/tmp/
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

scp remote to local

Copies a remote file into the current local directory.

```
$ scp ubuntu@10.0.0.5:/tmp/app.log .
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

scp -i

Copies a file over SSH using a specified private key.

```
$ scp -i key.pem app.log
ubuntu@10.0.0.5:/tmp/
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

OPTION CLUE

-i: ignore case or use an identity file, depending on the command

rsync

Synchronizes only changed files while preserving useful metadata.

```
$ rsync -av /app/
ubuntu@10.0.0.5:/backup/app/
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

OPTION CLUE

-a: include all or preserve metadata, depending on the command

rsync over SSH

Synchronizes files through SSH with a specified private key.

```
$ rsync -av -e 'ssh -i key.pem' /app/
ubuntu@10.0.0.5:/backup/app/
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

OPTION CLUE

-a: include all or preserve metadata, depending on the command; -i: ignore case or use an identity file, depending on the command

18

Log troubleshooting

7 command examples in this section

tail -f

Checks live system logs on Ubuntu.

```
$ sudo tail -f /var/log/syslog
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

tail -f

Checks authentication logs on Ubuntu.

```
$ sudo tail -f /var/log/auth.log
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

tail -f

Checks system logs on RHEL/CentOS.

```
$ sudo tail -f /var/log/messages
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

journalctl -u

Checks live logs of a service.

```
$ journalctl -u nginx -f
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

tail -f

Checks live application logs.

```
$ tail -f /app/logs/app.log
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

dmesg

Shows kernel logs.

```
$ dmesg
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

dmesg -w

Follows live kernel logs.

```
$ dmesg -w
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

19

Cron scheduling

5 command examples in this section

crontab -e

Edits cron jobs.

```
$ crontab -e
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

crontab -l

Lists current cron jobs.

```
$ crontab -l
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

Every 5 minutes

Runs the script at minute 0, 5, 10, and so on.

```
$ */5 * * * * /home/ubuntu/healthcheck.sh
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Daily at 2 AM

Runs the backup once a day at 02:00 server time.

```
$ 0 2 * * * /backup.sh
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Every Sunday

Runs the cleanup every Sunday at 01:00 server time.

```
$ 0 1 * * 0 /cleanup.sh
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

20

DevOps troubleshooting

15 command examples in this section

Check service status

Checks whether nginx is running or failed.

```
$ systemctl status nginx
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Restart service

Restarts a service.

```
$ sudo systemctl restart nginx
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

Check open ports

Shows listening ports.

```
$ ss -tulnp
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Check CPU usage

Shows live CPU usage.

```
$ top
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Check memory

Shows memory usage.

```
$ free -h
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-h: human-readable units

Check disk usage

Shows filesystem disk usage.

```
$ df -h
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-h: human-readable units

Check directory size

Shows size of `/var/log`

```
$ du -sh /var/log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Check live logs

Follows live system logs.

```
$ tail -f /var/log/syslog
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-f: follow output or force an action, depending on the command

Find process

Finds Java process.

```
$ ps aux | grep java
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

Kill process

Forcefully kills a process.

```
$ kill -9 <PID>
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

SAFETY

Caution: SIGKILL prevents cleanup. Try a normal kill first and use -9 only when necessary.

Test API

Calls a health endpoint to verify that the application responds.

```
$ curl http://localhost:8080/health
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Check port usage

Shows which process has port 8080 open.

```
$ sudo lsof -i :8080
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-i: ignore case or use an identity file, depending on the command

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

Check reboot time

Displays the time of the most recent system boot.

```
$ who -b
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Check login history

Reads the login database and shows recent user sessions.

```
$ last
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Watch output

Reruns disk usage every two seconds and refreshes the screen.

```
$ watch df -h
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-h: human-readable units

21

Useful one-liners

9 command examples in this section

Largest files

Finds files, calculates sizes, sorts largest first, and keeps the top ten.

```
$ find / -type f -exec du -h {} +
2>/dev/null | sort -rh | head -n 10
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-h: human-readable units; -r: recursive or reverse, depending on the command

Top memory processes

Sorts the process list by memory use and shows the first entries.

```
$ ps aux --sort=-%mem | head
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

Top CPU processes

Sorts the process list by CPU use and shows the first entries.

```
$ ps aux --sort=-%cpu | head
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

Listening TCP ports

Lists listening TCP sockets with numeric addresses and process details.

```
$ sudo ss -ltnp
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

Count directory entries

Places one name per line and counts the lines.

```
$ ls -l | wc -l
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

Search system errors

Searches syslog for error in any letter case.

```
$ grep -i 'error' /var/log/syslog
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-i: ignore case or use an identity file, depending on the command

Archive logs

Creates a gzip-compressed archive of the log directory.

```
$ tar -czvf logs-backup.tar.gz /var/log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Old log files

Finds .log files modified more than seven 24-hour periods ago.

```
$ find /var/log -name '*.log' -mtime +7
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-n: numeric output or a requested count

Largest log items

Measures log items, sorts by size, and shows the ten largest.

```
$ du -ah /var/log | sort -rh | head -n 10
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-a: include all or preserve metadata, depending on the command; -r: recursive or reverse, depending on the command

22

Interview essentials

27 command examples in this section

ls -la

Lists all files with details.

```
$ ls -la
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-l: long/detailed listing

cd

Changes directory.

```
$ cd /etc
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

pwd

Shows current directory.

```
$ pwd
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

cat

Displays file content.

```
$ cat file.txt
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

less

Opens large files page by page.

```
$ less app.log
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

head

Shows first 10 lines.

```
$ head app.log
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

tail -f

Follows live logs.

```
$ tail -f app.log
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

grep

Searches text in files.

```
$ grep error app.log
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

find

Finds files.

```
$ find / -name file.txt
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE

-n: numeric output or a requested count

chmod

Changes permissions.

```
$ chmod +x script.sh
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

chown

Changes ownership.

```
$ chown user:group file.txt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

ps aux

Shows running processes.

```
$ ps aux
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

top

Shows live system usage.

```
$ top
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

kill

Kills a process.

```
$ kill -9 PID
```

WHEN TO USE

Use when diagnosing or controlling running programs and shell jobs.

SAFETY

Caution: SIGKILL prevents cleanup. Try a normal kill first and use -9 only when necessary.

df -h

Checks disk usage.

```
$ df -h
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

OPTION CLUE

-h: human-readable units

du -sh

Checks directory size.

```
$ du -sh /var/log
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

free -h

Checks memory.

```
$ free -h
```

WHEN TO USE

Use for a quick capacity or resource check before taking corrective action.

OPTION CLUE

-h: human-readable units

ip a

Shows IP address.

```
$ ip a
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

ping

Checks connectivity.

```
$ ping google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

curl

Tests HTTP/API endpoint.

```
$ curl
```

WHEN TO USE

Use to test a URL, inspect an HTTP response, or retrieve content without a browser.

ss -tulnp

Checks open ports.

```
$ ss -tulnp
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

systemctl

Checks service status.

```
$ systemctl status nginx
```

WHEN TO USE

Use on systemd-based Linux distributions to inspect or control a service unit.

journalctl -u

Checks service logs.

```
$ journalctl -u nginx -f
```

WHEN TO USE

Use during troubleshooting when you need recent or continuously arriving log messages.

OPTION CLUE

-f: follow output or force an action, depending on the command

ssh

Connects to remote server.

```
$ ssh ubuntu@server-ip
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

scp

Copies files between systems.

```
$ scp file.txt ubuntu@server:/tmp/
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

tar

Compresses files/directories.

```
$ tar -czvf backup.tar.gz /app
```

WHEN TO USE

Use when packaging files for backup, transfer, or extraction.

crontab -e

Edits scheduled jobs.

```
$ crontab -e
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

23

Scenario playbook

15 command examples in this section

Application is down

Checks whether the service failed, exited, or is still running.

```
$ systemctl status app-service
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Need application logs

Streams new log entries for the service while you reproduce the issue.

```
$ journalctl -u app-service -f
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-f: follow output or force an action, depending on the command

Disk is full

Identifies which mounted filesystem has little or no free space.

```
$ df -h
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-h: human-readable units

Find large files

Finds the largest paths; expect permission errors unless run with suitable access.

```
$ du -ah / | sort -rh | head -n 10
```

WHEN TO USE

Use when you need to filter, locate, transform, or summarize text from files or another command.

OPTION CLUE

-a: include all or preserve metadata, depending on the command; -r: recursive or reverse, depending on the command

High CPU

Shows live CPU load and the processes currently consuming it.

```
$ top
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

High memory

Ranks processes by memory percentage to identify likely offenders.

```
$ ps aux --sort=-%mem | head
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Port conflict

Identifies the process listening on or connected through port 8080.

```
$ sudo lsof -i :8080
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

OPTION CLUE

-i: ignore case or use an identity file, depending on the command

SAFETY

Runs with elevated privileges. Read the full command and verify every path before pressing Enter.

Network issue

Checks basic reachability and round-trip latency; some hosts block ping.

```
$ ping google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

DNS issue

Tests whether a name resolves to an address through the configured DNS server.

```
$ nslookup google.com
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

API not responding

Requests only response headers to check status and server reachability.

```
$ curl -I http://localhost:8080
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.

Permission issue

Inspects owner, group, and permission bits before changing anything.

```
$ ls -l file.txt
```

WHEN TO USE

Use after inspecting the target and confirming the change is intended.

OPTION CLUE

-l: long/detailed listing

Script will not run

Adds execute permission; also verify the script interpreter and ownership.

```
$ chmod +x deploy.sh
```

WHEN TO USE

Use when narrowing a network problem from local configuration to reachability, ports, or DNS.

SSH issue

Enables verbose connection diagnostics for keys, host checks, and authentication.

```
$ ssh -v user@server-ip
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

OPTION CLUE

-v: verbose output

Failed SSH logins

Finds failed authentication attempts on Debian-based systems.

```
$ grep 'Failed password' /var/log/auth.log
```

WHEN TO USE

Use for secure remote administration or file transfer; verify the host identity before trusting it.

Recent logs

Limits the journal to entries created during the last hour.

```
$ journalctl --since '1 hour ago'
```

WHEN TO USE

Use this when the described result helps you inspect or manage the current Linux system.